המכון למחקרי נפט וגיאופיסיקה
THE INSTITUTE FOR PETROLEUM RESEARCH AND GEOPHYSICS

MAGNETOTELLURIC SURVEY

IN NORTHERN ISRAEL

By

Y. Rotstein

S. Goldberg

Prepared for

OFFICE OF THE CHIEF SCIENTIST

MINISTRY OF COMMERCE, INDUSTRY AND TOURISM

MARCH 1979

MT/325/77

MAGNETOTELLURIC SURVEY

IN NORTHERN ISRAEL

By

Y. Rotstein

S. Goldberg

Prepared for

OFFICE OF THE CHIEF SCIENTIST

MINISTRY OF COMMERCE, INDUSTRY AND TOURISM

MARCH 1979

MT/325/77

TABLE OF CONTENTS

	Page
INTRODUCTION	1
FIELD PROCEDURE AND INSTRUMENTATION	2
DATA ANALYSIS AND INTERPRETATION	3
RESULTS AND DISCUSSION	4
General	4
Upper Galil	4
Golan Heights	5
Hula Valley	7
Crustal Resistivity Structure	9
CONCLUSIONS AND RECOMMENDATIONS	12
REFERENCES	14

INTRODUCTION

This report describes the results of seven out of ten magnetotelluric soundings carried out in northern Israel by the Institute for Petroleum Research and Geophysics. The survey was made on behalf of the Office of the Chief Scientist, Ministry of Commerce, Industry and Tourism. Three additional soundings proved to be noisy and did not carry any useable information. Also included in this report is a discussion of one sounding carried out on behalf of TAHAL (Rotstein and Goldberg, 1977).

The present survey was conducted for two reasons. The first being to try and collect more data on the subsurface structure and lithology and the second, was to examine the MT response of different provinces in northern Israel. Since the MT method's ability to supply reliable data in a complex geological province is still uncertain, the present survey was conducted on a limited scale.

The location of the various soundings, including the unsuccessful ones, are shown in Figure 1. The surveyed area includes three geomorphological provinces: Upper Galil, Hula Valley and the Golan Heights.

The Upper Galil is a mountainous region with mountains as much as 1200 m high and with deep valleys between them. Much of the topography is controlled by a series of normal faults which intersect the area in various directions. Particularly prominent are the E-W faults that create a series of valleys in which the successful soundings were carried out. Exposed rocks in this area are mostly upper Cretaceous carbonates. The complex geological structure of the area makes it difficult to locate suitable MT sites. In general, an effort was made to situate the soundings away from obvious major structural and lithological discontinuities but this was not always possible.

The easternmost part of the surveyed area, the Golan Heights, is a basalt covered plateau. The basalt is of Tertiary age and originated in a series of volcanic cones and fissures near the Syrian-Israeli border. Carbonate rocks, much like the rocks in Upper Galil, appear beneath the surface volcanics and form a large syncline (Michelson, 1972). Since thick volcanic flows cover most of the area, not much is known about the structure of the underlying sedimentary rocks. The young volcanism and the area's proximity to the Dead Sea Rift have long made it a prime target for geothermal exploration. A D.C. resistivity survey (Rotstein et. al., 1978) conducted near the volcanic cones have shown that no geothermal reservoirs can be expected in the subsurface and that no residual heat is associated with the volcanic activity.

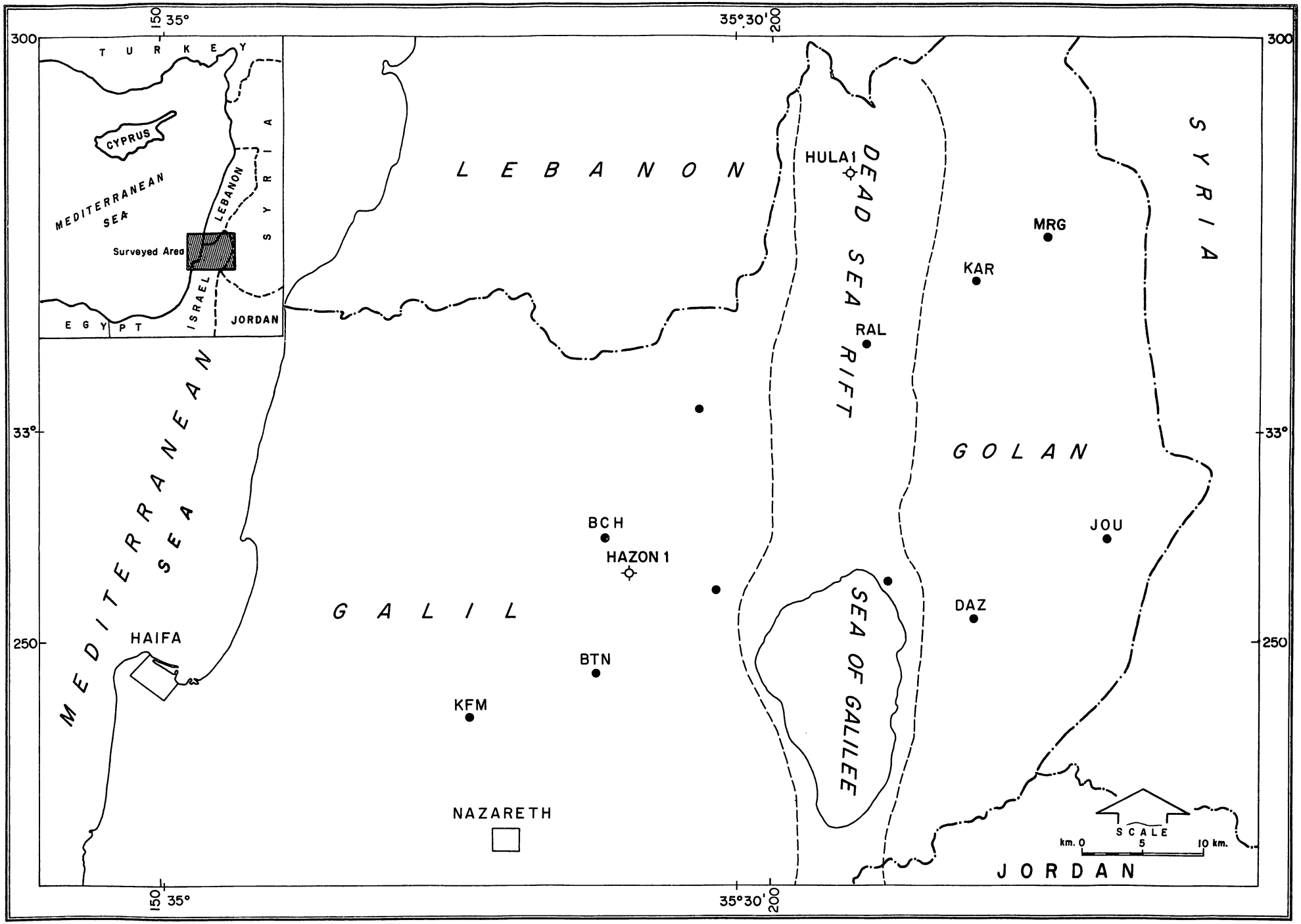


Figure 1 Location of MT soundings in Northern Israel.
Unnamed sites represent soundings with noisy data.

MT/325/77(55993)

Between the Golan Heights and Upper Galil appears a geomorphologically low area, marking that part of the Dead Sea Rift lying in this region (Fig. 1). High angle normal faults seem to mark the boundaries of the rift in this area but they are not always exposed to the surface. The difference in elevation between the rift and the elevated regions on both sides is usually in the order of several hundred meters. In Lake Kinneret, the rift is at an elevation of about -200 m while the elevated areas nearby reach more than 500 m. Basically, the rift is a two dimensional structure trending N-S; however, the structure is by no means uniform along the rift axis and local structures are known to exist at various places along the rift. The Hula Valley, which appears to be a structural depression by its gravity signature (Yuval, 1960; Folkman, 1977), is one of these local structures.

Traditionally, the MT method has been used in the structural analysis of gross features (mostly layering) in sedimentary provinces. Ideally, it is used only in very simple geological structures; preferably, those which are basically one dimensional (e.g. - layered cake geology). In a more complex geology, use of the method is not as favourable and the interpretation may get troublesome since no routine and reliable interpretation schemes have yet been developed for two or three dimensional cases. However, experience has shown that it is not always easy to decide beforehand whether the geoelectrical structure in an area is complex or not. In the Upper Galil, for example, the geological complexity is obvious already on the surface. On the other hand, some of the different geological units present in the area, both on the surface and in the subsurface, have rather similar geoelectrical characteristics. Consequently, the area's response to an MT sounding is difficult to predict.

FIELD PROCEDURE AND INSTRUMENTATION

A five-channel digital magnetotelluric system manufactured by Geotronics Inc. was used in this survey. Changes in the magnetic field were recorded via induction coils. Cd-$CdCl_2$ non-polarized electrodes were used to detect the telluric current. At a few of the sites, the non-polarized electrodes were replaced by lead plates which proved easier to handle and had no adverse effect on signal quality.

Recordings were made at a frequency range of about 40 to 0.002 Hz. For the purposes of effective gain control and effective sampling rate, this wide frequency interval was divided into six frequency bands. A number of runs,

each with 10^{12} - 10^{13} samples, were recorded at each of the four higher frequency bands. At the two lower frequency bands, only one run was recorded at each site. A sampling rate of 10 sec was used for a typical run at the lowest frequency and with 10^{12} samples, lasting about 11½ hours. These long recordings were usually done overnight with no operator present.

Finding suitable site locations for the present survey proved to be a difficult task due to the combination of rugged topography, complex geology, cultural noise and restricted military areas. In the Upper Galil where site location was especially difficult, successful sites were placed in the East-West structurally controlled valleys, such as Bet Netofa and Bet haKerem.

DATA ANALYSIS AND INTERPRETATION

The field tapes were analysed following a procedure described by Sims et. al. (1971). Tensor apparent resistivity values were rotated to their principal directions - corresponding to the geoelectric strike and perpendicular to it (These directions are hereon refered to as TE and TM). Phases for each of the apparent resistivity components were also computed and used in the analysis. Data with phasor coherencies (Sims et. al., 1971) of 0.9 and better were used in the interpretation except for occasional instances where data with coherencies of 0.8-0.9 were judiciously added. These lower coherencies were added only to supplement existing higher coherency values at the same frequencies, and when they did not generate any irregular complexity to the data.

For interpretation, we selected the TE curves in a one dimensional inversion. Such an approach is fully justified in a one dimensional structure and reasonably justified in a two dimensional structure with mild dips. Under these conditions, Patrick and Bostick (1969) and Reddy and Rankin (1972) have shown that a one dimensional inversion of the TE data yields a resistivity cross-section which satisfactorily approximates the true resistivity section found under the site. This approach was successfully implemented by various investigators under similar conditions (e.g. Word et. al., 1970; Vozzof, 1972; Reddy and Rankin, 1972; Rotstein and Goldberg, 1977; Rotstein et. al., 1978; and Shoham and Ginzburg, 1978). In the present survey, however, only a few stations in the Golan Heights and Hula Valley may possibly be regarded as complying to these restrictions. At other stations, particularly in the Galil and Golan Heights, the structure cannot be regarded as one dimensional or two dimensional with mild dips. In such cases, the one dimensional inversion becomes less reliable and more problematic; even if

the available field data is of the best technical quality.

RESULTS AND DISCUSSION

General

In Figures 2 and 3, the results of the eight soundings are shown with their subsurface interpretations. At each site, the apparent resistivity and phase data are plotted for the two components; one in the direction of the geoelectric strike (TE) and the other, in a direction perpendicular to it (TM). In most of the soundings, the TE and TM data follow distinct and different curves which are usually far enough apart to allow us to consider the structure as not being one dimensional.

For the sake of simplicity and coherency, we have divided the description and discussion of the results into two parts. The first deals with the analysis of higher frequencies in the different soundings in terms of the sedimentary structure. The second discusses the application of the lower frequency portion of these soundings to the study of the crust and upper mantle in Israel.

Upper Galil

Three of the soundings were recorded in the Upper Galil. All of these soundings exhibit a strong tendency to deviate from a one dimensional case (Fig. 2). In all of them, TE and TM data is different over most of the recorded spectrum. It is interesting, however, that in the two soundings that include detailed TE data (KFM and BCH; Fig. 1, 2), the basic features of the apparent resistivity are similar.

1) TE is lower than TM over all of the recorded spectrum.
2) TE data shows a basic division of the sedimentary section into four resistivity layers.
3) None of the soundings indicate any higher resistivity basement.

Of the resistivity layers within the sedimentary sequence, the second and the third units can best be compared with the available subsurface information. The second unit, which follows the low resistivity surface layer, seems to have a thickness of 450-700 m. The third, with lower resistivity extends to a depth of 1400-1900 m below the surface at BCH and KFM. At this depth, the resistivity increases but the detailed resistivity structure of the fourth layer is not resolved. The only layers, possibly present, which can explain the large resistivity

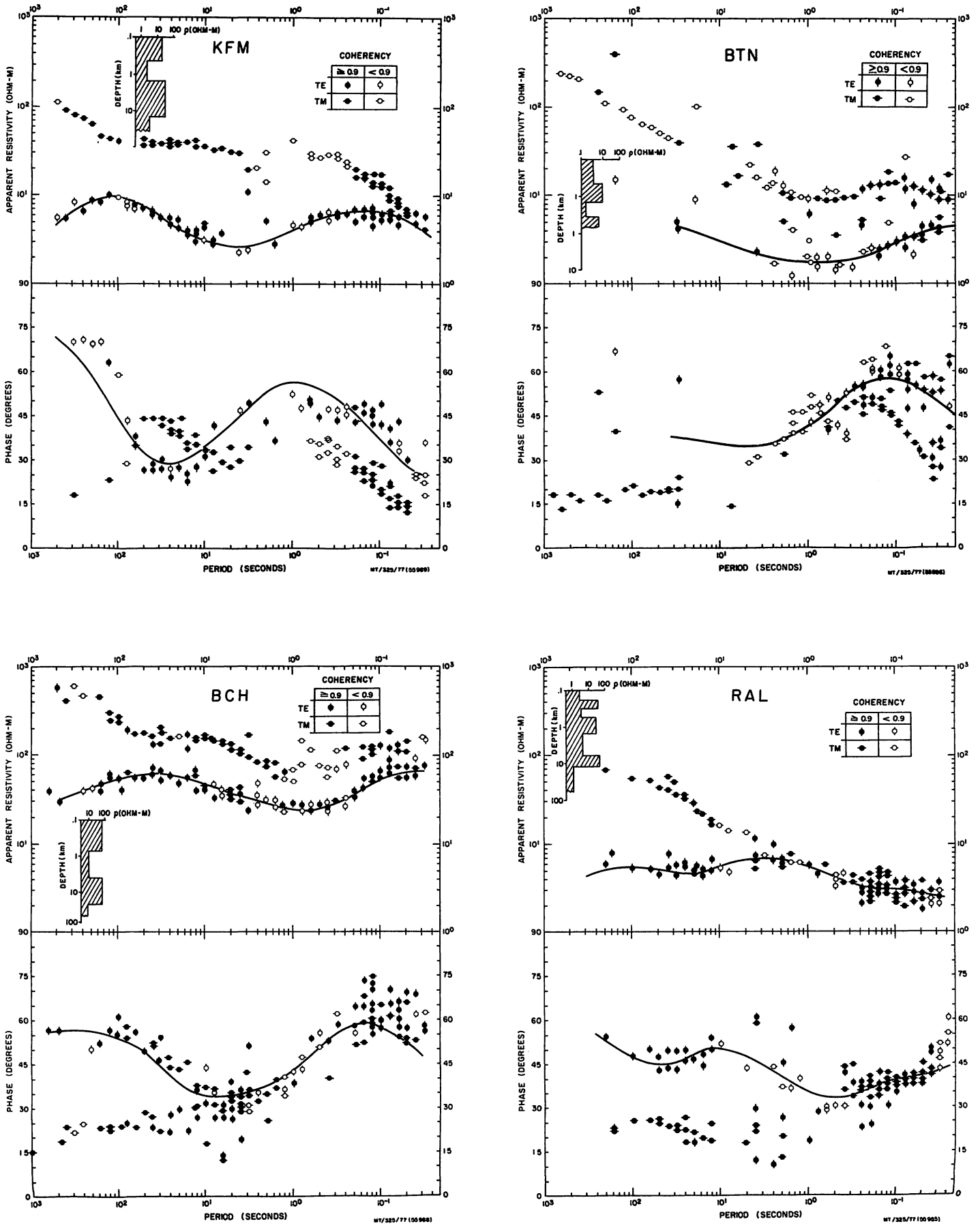


Figure 2 Apparent resistivity, phase and inverted subsurface resistivity models for MT soundings in the Galil and Hula Valley.

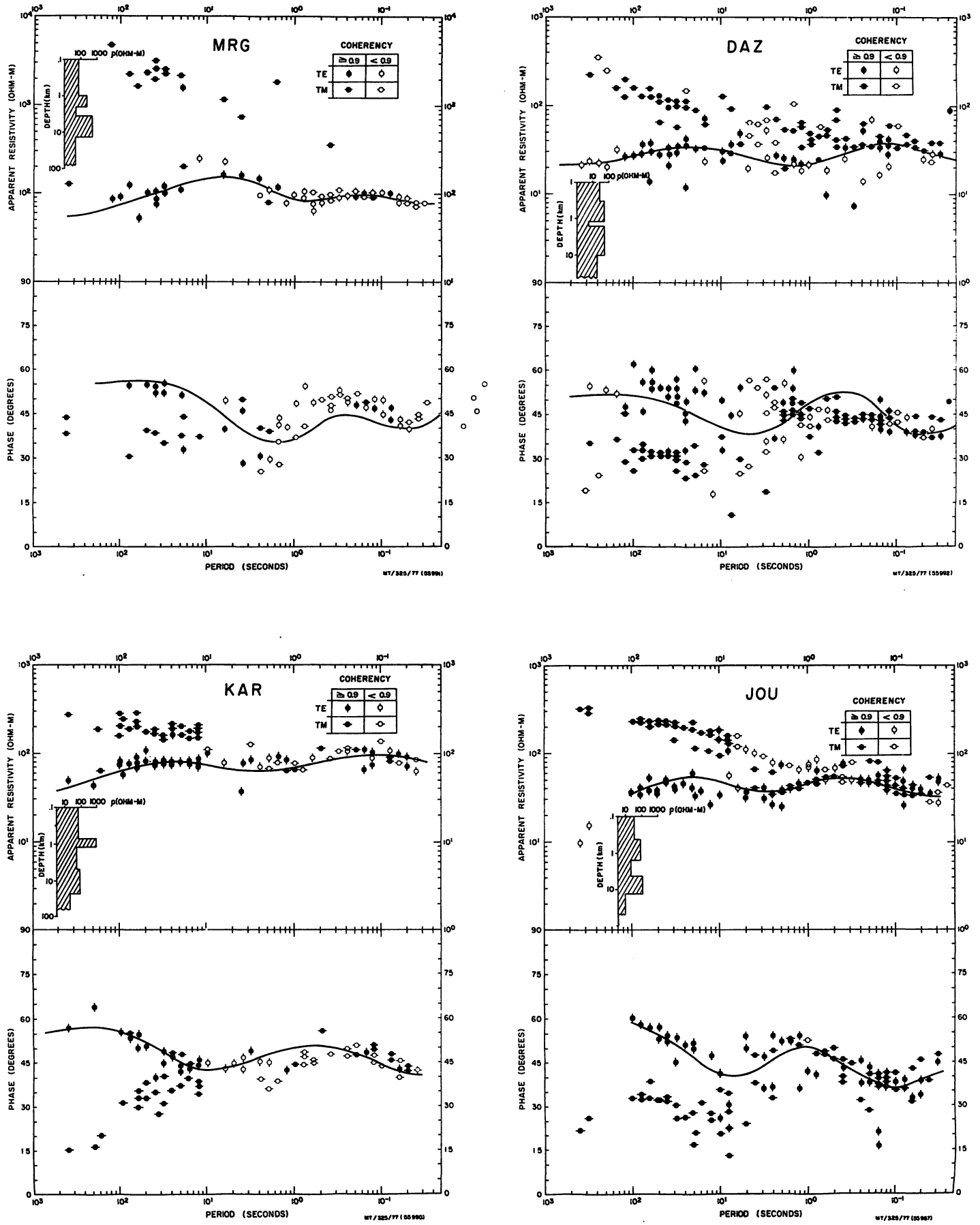


Figure 3 Apparent resistivity, phase and inverted subsurface resistivity models for MT soundings in the Golan Heights.

increase at a depth of 1400-1900 m are the Jurassic limestones encountered in the Hazon 1 well (Fig. 1) at a depth of about 1100 m, and in the Sarid 1 well at a depth of about 1650 m. Lower Cretaceous limestones, shales and sandstones which overlie the Jurassic limestones invariably have lower resistivity. This is clearly indicated in MT surveys in southern Israel (Shoham and Ginzburg, 1978). These rocks, therefore, can correspond with the relatively low resistivities observed in the soundings at depths of over 450-700 m. The higher resistivities observed above 450-700 m can only be attributed to the limestone-dolomite sequence of the Judea group.

One limitation of the MT method in this area is its inability to resolve into layering the high resistivity found at depths of over 1400-1900 m. The upper part of this high resistivity may, possibly, correspond to the Jurassic limestones, but the thick sedimentary sequence underlying the Jurassic rocks is not clearly indicated in the data.

Another drawback to this method stems from the fact that the derived layer solution, reasonable as it may sound, is somewhat speculative in this region due to the nonhomogeneity of the subsurface. In effect, it is very likely that although all the described units are present, the thicknesses which were determined for them are not accurate. The expected error may easily be in the range of several tens of a percent of the total depth to the respective layer.

Golan Heights

Four soundings made in the Golan Heights (MRG, KAR, JOU, DAZ; Fig. 3) are described in this work. The quality of the data is good in most of the soundings. Therefore, the reliability of the inversion into a subsurface resistivity structure depends mostly on the resolution of the method. Of the three areas surveyed, soundings in the Golan Heights show the highest degree of homogeneity, as evidenced by the separation of TE and TM data. In effect, sounding KAR and to some degree JOU (Fig. 3) show a deviation from a one dimensional appearance only at the low frequencies. These results are not surprising since the area is covered with several hundred meters of basalt flows, which are basically a one dimensional structure.

The inverted subsurface structure of the TE data from the various soundings is presented in Figure 3. In all of the soundings, there is a basic division of the sedimentary section into four resistivity layers. Even though a quantitative inversion was performed for all four soundings, the more meaningful ones are KAR

and JOU. The subsequent discussion will focus only on these two soundings. The sedimentary section in the Golan Heights seems to be resolved into the following resistivity layers.

The first layer is near the surface and has resistivities ranging from 25 to 80 ohm-m. The layer's thickness appears to be in the range of 320 to 1000 m. In soundings JOU and KAR, which appear to be the most reliable, this layer is 470-700 m thick and appears to have a resistivity of 32-75 ohm-m. Considering the fact that the area is covered with high resistivity basalt flows, the appearance of a lower resistivity, near surface layer may seem surprising. A possible explanation, however, is that the first resistivity layer is in effect a combination of the volcanic series and some low resistivity sedimentary formation underneath, which in turn reduces the effective resistivity of the sequence. An alternative but closely related explanation is that the high resistivity, near surface volcanics are not seen because the high frequency cut-off of the data is too low; therefore, only the low resistivity, upper layer of the sedimentary sequence is observed.

One indication of the validity of these explanations can be seen in a D.C. survey made in the northeastern Golan Heights (Rotstein et. al., 1978). There, a sounding near the JOU, MT site shows that the high resistivity volcanics are underlaid by a low resistivity layer. This lower resistivity layer underlying the basalt is well known in the southern Golan Heights. In this area, Eocene chalks and chalky limestones (Michelson, 1975) make up most of the sequence directly beneath the basalts. In the northernmost Golan Heights, the Eocene sequence includes less chalk and is composed mostly of limestones (Michelson, 1975); there is no information as to where this change takes place.

The second, a higher resistivity layer with a resistivity of probably over 80 ohm-m, seems to have a thickness of about 500 m at KAR and 1600 m at JOU. In KAR, the appearance of this layer beneath 700 m of Tertiary basalts and possibly Eocene chalks and limestones may be correlated with the upper Cretaceous dolomites and limestones. These dolomites and limestones were determined by Michelson (1975) to be about 450 m thick near Birkat Ram. The much larger thickness of this layer in the JOU sounding is more difficult to explain.

The third is a lower resistivity layer with a resistivity of about 20-50 ohm-m. This resistivity layer includes all, or much of the rest of the sedimentary sequence from lower Cretaceous to the basement. The most reasonable estimate of rock composition in this sequence is that of both high (limestones, dolomites) and low (shales,

marls, sandstones) resistivity rocks. Lower Cretaceous rocks are known to appear as a lower resistivity layer in the Upper Galil and various other parts of Israel. Since the Jurassic sequence is known to be about 2-3 km thick (Freund, 1978) and mostly consists of high resistivity limestone, the low resistivity may reflect the presence of saline water.

Underlying these three resistivity layers is a fourth layer with a resistivity much higher than that of the previous one. Even though definition of the resistivity of this layer is not accurate, the fact that its resistivity increases drastically compared with the previous layer and the large depth to its top, both make it probable that these are basement or near basement rocks. Beneath this interface and to a depth of over 10 km, there is no indication of any additional change in resistivity. As before, the best definition of the depth to this layer can be derived from the KAR and JOU soundings, where it is estimated to be 4300-4900 m. This result is similar to the result of D.C. resistivity soundings (Rotstein et. al., 1978); in which depth to the high resistivity layer was determined to be about 4700-5400 m. Taking an average value of the results of the two methods, the high resistivity (basement?) in the Golan Heights can be estimated to be at a depth of about 4700 m below the surface.

It is also important for us to look at the results of the other two soundings, DAZ and MRG. In these soundings, a wide separation between TE and TM data marks the anisotropy and inhomogeneity of the subsurface. Again, under these conditions the general features of the subsurface are preserved in the TE data but depths to the different interfaces are distorted. The DAZ curve is probably the most distorted, having widely separated TE and TM data already in the higher frequencies.

The difference between the two groups of soundings; one which is reliable for quantitative interpretation and the other which is not, clearly indicates some of the problems associated with MT soundings. In particular, it demonstrates the difficulty in predicting the MT response of a region before performing several reconnaissance soundings.

Hula Valley

The Hula Valley is a small trough filled with clastic sediments within the much larger regional structure of the Dead Sea trough. Folkman (1976), based on gravity data, has estimated the depth to the dense and high resistivity layer at the bottom of the valley to be about 2000 m. Considering the gravity anomaly to be localized and only a few kilometers to the south, these same rocks are only

several hundred meters deep. We must conclude, therefore, that the Hula Valley is a distinct local structure and may have a three dimensional response to MT soundings.

The only MT sounding performed in the area (RAL) is shown in Figure 2 together with its inverted subsurface structure. Almost surprisingly, the TE and TM data seem to coincide over much of the recorded spectrum, particularly in the upper frequencies. This seems to indicate that the sounding is favourable for quantitative analysis.

The subsurface resistivity model includes a sequence of resistivity lows and highs. The first layer from the surface is 170 m deep and has a resistivity of about 3 ohm-m. The corresponding layer is known from the various wells drilled in the valley (Agron and Fleisher, 1976) to consist of clay, marl and peat. The high resistivity observed at a depth between 170 to 340 m may be explained by the presence of a basalt layer. The presence of such a layer was indicated a few kilometers to the north in H-1 well (Fig. 1), where a basalt layer is encountered at a depth of 200 m and extends to the bottom of the well, at about 260 m. At a depth of 340 m and up to a depth of 530 m, the resistivity drops again to about 3.3 ohm-m. A similar division of the clastic series into upper and lower portions, separated by a volcanic layer, was observed in a few other shallow wells in the valley (Agron and Fleisher, 1976). At a depth of 530 m, the resistivity increases up to a depth of about 1500 m. The most reasonable explanation for this sequence is the presence of the low resistivity clastic layer to a depth of 530 m and a limestone-dolomite sequence (Judea Group?) between 530 m to 1500 m. Another low resistivity sequence appears below and up to a depth of about 6500 m. This sequence includes the lower Cretaceous, Jurassic and Triassic rocks, apparently, with a large percentage of clays and marls which significantly reduce the effective resistivity of the sequence. The lower parts of the sedimentary sequence appear similar to it at other parts of this survey and the northern Negev.

A depth of 6500 m, determined for the high resistivity at the bottom of the sedimentary sequence, is in line with the 4500-5000 m depth observed for the same layer in the Golan Heights. The difference can be attributed to the vertical throw on the rift faults and this can be an indication of the amount of subsidence in the rift.

Similar to the Golan Heights, the large depth to the high resistivity rocks indicate that these are either basement or near basement rocks and that the

Paleozoic section appearing in the northern Negev as a thick high resistivity layer (Rotstein et. al., in preparation) is either small or completely missing.

Crustal Resistivity Structure

The MT method is widely used throughout the world for the study of crustal and upper mantle resistivity structure. It is particularly suitable for such research because large depth of penetration can be achieved with very short lateral spreads and simple logistics. This is unlike the D.C. resistivity method or seismic refraction, either of which can probe to great depths but involve very cumbersome field operations.

Numerous deep MT soundings have been carried out in North America and Europe (particularly in the Soviet Union), and a few in other parts of the world. Local measurements over a variety of terrains and upper crust geology reveal, as expected, that the upper few kilometers vary greatly in resistivity according to local geology. However, a large number of deep MT surveys exhibit a remarkable and unexpected phenomenon. It turns out that at a depth of over 10 Km but within the crust or uppermost mantle, a marked reduction in resistivity occurs. Table 1 (Fournier, 1978) includes a list of areas in which this resistivity low has been observed and approximate depths at which they appear. As can be seen in the list, the low resistivity layer is worldwide, but the depth to its top varies greatly. In all of these cases, the low resistivity does not appear in the base of the lithosphere where it is known to exist due to the partial melting of the upper mantle. Instead, it appears within the lithosphere which was thought until recently to be one uniform, high resistivity layer.

The same phenomenon was observed in all the MT surveys performed in Israel and the Sinai Peninsula (e.g. Shoham and Ginzburg, 1978; Rotstein et. al., 1978). It is clear, therefore, that the crust beneath Israel and nearby areas includes a low resistivity layer with an unknown thickness. In northern Israel and the Golan Heights, this layer appears to begin in the upper crust (Table 2) and only in two soundings, KAR and BCH, it appears at a depth greater than 20 Km.

Table 1

Country	Region	Depth to Top	Depth to Bottom
Germany	Geöttingen	15	30
Spain	Toledo	30	72
	Tortosa	37	60
France	Nozay	40	50
	Basin of Paris	35	45
	Mortaque Vendie	15	25
	Cher	35	55
Scotland	Midland Valley	12	
	Southern Uplands	24	
Australia	Southern Australia	5	
U. S. A.	Western U.S.A.	20-25	
	Western Texas	20	
	Wisconsin	8-15	
Canada	Western Canada	10 (?)	
Kenya	Gregory Rift	8	
	Kenyan Rift Valley	12	
Ethiopia	Afar	15	
Soviet Union	South Caspian Depression	40-60	
	Yakutiv	15-23	
	Tikhenya Bay	10-25	
	Lake Baikal	40-100	
	Kamchatka	40	90
	Voronezh Massive	17-29	
	Middle Russian Depression	8-20	

Table 1 Known low resistivity zones in the crust and uppermost mantle (within the lithosphere) at various locations around the world. Modified from Fournier, 1978.

Table 2

MT Sounding	Depth (Km)
RAL	13
KAR	23
BCH	22
MRG	14
JOU	13
DAZ	12
KFM	16

Table 2 - Depth to the top of the low resistivity crustal layer in northern Israel.

The large depths to the top of the low resistivity zone recorded at KAR and BCH are probably in error and can again be best explained by the effects of lateral inhomogeneity in resistivity; particularly so in BCH which exhibits large deviation from one dimensionality over all of the recorded spectrum. The explanation in sounding KAR is less certain. Of interest and almost surprising is the uniformity of the depths to this layer; at most of the sites, it appears at depths of 12 to 16 Km. This uniformity is in great contrast with the large deviations in depths to the different sedimentary layers. These deviations were explained by the poor lateral homogeneity in the geology of the area. The important property of MT soundings exhibited, therefore, is that lateral disturbances in the one dimensionality at shallow depth do not effect the appearance of the deep structure to any large degree. This makes the MT method a very valuable tool in deep lithospheric investigations.

The physical explanation for this crustal low resistivity is not clear. One common explanation is that at these depths, the rocks consist of hydrated crustal or upper mantle material - possibly sepentinite (Van Zijl, 1977).

A second explanation was suggested by Shoham and Ginzburg (1978) to account for low crustal resistivity in southern Israel, relating it to a partially molten lower crust. This explanation, which may be true in some tectonically active areas, requires a high heat flow regime over the region since a partially molten crust and a normal heat flow are unacceptable. In Israel, however, it was clearly

shown by Eckstein (1976) that the heat flow over the entire country is, at the most, normal and probably less than the world average. Therefore, no explanation which assumes a partially molten lower crust or uppermost mantle can be regarded as proper.

Evidence of this widespread resistivity low is relatively new and research concerning this layer's distribution, thickness and physical meaning is still underway in many research centers around the world. The MT method is the most widely used method for this purpose.

CONCLUSIONS AND RECOMMENDATIONS

The present survey had a two-fold purpose. It was aimed at trying to follow, in a NE-SW profile, changes in the sedimentary section and crustal structure across northern Israel. In addition, this survey being the first MT survey to be performed in northern Israel, was to evaluate the use of the MT method in complex geological structures such as those present in the region.

The first aim was only partially fulfilled because the complexity of the region made it impossible to receive an accurate subsurface resistivity model at all places. In the Upper Galil, some possible correlation between known geology and the results of this survey was observed; but the exact subsurface structure is somewhat uncertain due to the large apparent resistivity anisotropy which is observed in the soundings. Sounding RAL in the Hula Valley seems to be favorable to one dimensional inversion. The derived subsurface model includes a succession of high and low resistivity layers which may add to the known data for the region. Observed in the data are the presence of a high resistivity (volcanic ?) layer at depths of 170-340 m below the surface and the termination of the low resistivity clastic series at a depth of about 530 m. The high resistivity basement in the rifted valley is observed about 6500 m below the surface.

In the Golan Heights, the high resistivity basement is probably about 4700 m below the surface; as expected, this is not as deep as in the rift where subsidence took place. The sedimentary sequence in the Golan Heights is divided into three resistivity layers. The correlation of these layers with the known lithology is not clear since no deep wells have been drilled in the region.

The results of this survey show the MT method to be significantly meaningful in the investigation of the sedimentary section only at certain locations in northern Israel.

In the Upper Galil, chances are that only a few MT soundings will ever yield

meaningful quantitative results. In the rift valley and in the Golan Heights, the use of the method is more appropriate. However, in these regions too, only a portion of the recorded soundings are expected to yield data which will fit for quantitative inversion.

The use of the MT method in these areas should be considered in light of the significance of the results. If the soundings have an important and immediate application, such as the location and planning of well drilling, a limited MT survey may be justified. If, however, a survey which has no immediate practical application is being considered, the MT method should be avoided in the Galil and should be used with caution in the rest of the area since only partial results are expected.

The MT method has repeatedly proved itself to be a very effective tool in crustal and upper mantle investigations. In this method, non-homogeneity near the surface does not severely effect the low frequency data in most cases. The MT method is widely used in deep surveys around the world and should be used for this purpose in Israel.

In northern Israel, the different soundings show a marked decrease in resistivity at depths of over 12-16 Km. The physical explanation for this low resistivity is not yet resolved.

REFERENCES

Agron, N. A. and Fleisher, E., 1976. Quality and Distribution of Peat in the Hula Valley (Interim Report), Geological Survey of Israel, HYDRO/1/76.

Eckstein, Y., 1977. The Measurement and Interpretation of Terrestrial Heat Flow in Israel, Ph.D. thesis, Hebrew University, Jerusalem, Israel.

Folkman, Y., 1976. Magnetic and Gravity Investigations of the Crustal Structure in Israel, Ph.D. thesis, Tel Aviv University, Tel Aviv, Israel.

Fournier, H., 1978. Recent Work in Magnetotelluric Sounding of the Lower Crust and Upper Mantle, Proceedings of the 4th. Workshop on Electromagnetic Induction in the Earth and Moon, Murnau, Germany.

Freund, R., 1978. The Judean Hills and Galilee Regional Synthesis of sedimentary basins. Part 1, Guidebook, Tenth International Congress on Sedimentology.

Michelson, H., 1972. The Hydrology of the Southern Golan Heights. T.H.L. Rep. HR/72/037 (In Hebrew).

Michelson, H., 1975. The Geohydrology of the Enclave and the Southeastern Slopes of Mount Hermon. T.H.L. Rep. 01/75/05.

Patrick, F. W. and Bostick, F. X., 1969. Magnetotelluric Modeling Techniques, E.G.R.L. Technical Report, No. 59, University of Texas, Austin, Texas.

Reddy, I. K. and Rankin, O., 1972. On the Interpretation of Magnetotelluric Data in the Plains of Alberta, Can. J. Earth Sci., No. 9.

Rotstein, Y. and Goldberg, S., 1977. Magnetotelluric Soundings in Nahal Daliot, Southeastern Golan Heights (in Hebrew), Inst. for Pet. Res. and Geophys., Report no. MT/234/76.

Rotstein, Y., Goldberg, S. and Klang, A., 1978. Electrical Resistivity Survey in the Northeastern Golan Heights, Inst. for Pet. Res. and Geophys., Report no. GE/380/77.

Rotstein, Y., Goldberg, S. and Neev, D., 1978. Magnetotelluric Survey in the Northwestern Sinai Peninsula, Inst. for Pet. Res. and Geophys., Report no. MT/194/75.

Sims, W. E., Bostick, F. X., Jr. and Smith, H. W., 1971. The Estimation of Magnetotelluric Impedance Tensor Elements from Measured Data, Geophysics, Vol. 36, pp. 938-942.

Shoham, Y. and Ginzburg, A., 1978. Crustal Structure in Central Israel from the Inversion of Magnetotelluric Data, Jour. Geoph. Res., Vol. 83, No. B 9.

Van Zijl, J. S. V., 1977. Electrical Studies of the Deep Crust in Various Tectonic Provinces of Southern Africa, Geophysical Monograph 20, The Earth's Crust.

Vozzof, K., 1972. The Magnetotelluric Method in the Exploration of Sedimentary Basins, Geophysics, Vol. 37, pp. 98-141.

Word, D. R., Smith, H. W. and Bostick, F. X., Jr., 1970. An Investigation of the Magnetotelluric Impedance Tenzor Method, E.G.R.L. Technical Report, No. 82, University of Texas, Austin, Texas.

Yuval, Z., 1966. Jordan Valley Gravity Survey, Inst. for Pet. Res. and Geophys., Report no. GR/427/66.